

Cross-Chain Deterministic Identity Mapping Between XRPL and Stellar *Technical Note* — Version 1.0 (Leon Calvin Long II)

Objective Demonstrate how a dataset or asset-state can be represented by a deterministic 32-byte seed using standard cryptographic primitives, and how that seed can be anchored on both the XRP Ledger and the Stellar Network to provide cross-chain identity continuity.

1. Deterministic State Representation A dataset $\mathcal{D}$ (e.g., a document, receipt, or structured asset-state) is normalized and hashed using a collision-resistant function such as SHA-256:

$$\sigma = \text{SHA256}(\mathcal{D})$$

The output σ is a 32-byte seed that uniquely represents the dataset. Unlike compression, this seed is not reversible — reconstruction requires possession of the original dataset or a canonical tokenization layer.

2. Canonical Tokenization Layer (Optional) If deterministic reconstruction is required, the dataset can be encoded using a reversible canonical tokenization scheme. In this case, the seed σ acts as the identity anchor, while the canonical tokenization layer stores the reversible structure.

3. Cross-Chain Anchoring The seed σ can be anchored on both chains using native mechanisms:

- **XRPL:** Stored in a transaction memo, account domain field, or emitted as part of an x-chain proof.
- **Stellar:** Stored in a ManageData entry or embedded in a contract invocation.

Anchoring the same seed on both chains establishes deterministic identity continuity:

$$\sigma_{\text{XRPL}} = \sigma_{\text{Stellar}} = \sigma$$

4. Parity Condition If both chains store the same seed, then any system performing verification can assert:

$$\text{Verify}(\sigma, \mathcal{D}) = \text{true}$$

This provides a cross-chain identity invariant without requiring wrapped assets or synthetic representations.

5. Reconstruction

- If canonical tokenization is used:

$$\Gamma(\sigma, \text{canonical_tokens}) \rightarrow \mathcal{D}$$

- If only hashing is used: Reconstruction requires the original dataset; the seed only verifies integrity.

6. Security Considerations

- Use standard cryptographic primitives (SHA-256, SHA-512, Blake2).
- Avoid proprietary or unverifiable algorithms.
- Anchors should be immutable and publicly verifiable.
- Cross-chain parity should be validated through independent observers or oracles.

Conclusion A 32-byte deterministic seed can serve as a cross-chain identity anchor between XRPL and Stellar using standard cryptographic methods. This provides a simple, secure, and interoperable way to maintain identity continuity across heterogeneous ledgers without requiring wrapped assets or synthetic state